Regime-Dependent Predictive Structure Between Equity Factors: Evidence from Granger Causality

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Abstract

We document regime-dependent predictive structure between equity factors. Using 35 years of Fama-French data (1990–2024), we find that Value (HML) Granger-causes Size (SMB) during crisis regimes ($p < 10^{-4}$, 9-day lag), but not during normal conditions. This pattern validates across 5 of 6 historical stress events (2008, 2011, 2015, 2018, 2020). We identify regimes using a Student-t HMM, which detects moderate crises (2011: 69%) that Gaussian models miss entirely (0%). While this relationship does not generate trading profits, the 9-day lead time may inform risk management decisions. We emphasize that Granger causality establishes temporal precedence, not structural causality—common drivers could explain the pattern. Our economic interpretation is a hypothesis, not a verified mechanism.

CCS Concepts

• **Mathematics of computing** → **Time series analysis**; • **Computing methodologies** → **Causal reasoning and diagnostics**.

Keywords

Factor Investing, Granger Causality, Regime Switching, Hidden Markov Models, Risk Management

1 Introduction

The August 2007 quantitative meltdown, in which systematic equity strategies lost approximately 30% in three days [7], revealed a critical blind spot in factor-based risk management. When multiple quantitative funds held similar factor exposures, forced liquidation by one fund created price pressure that cascaded to all others. Standard correlation-based risk models failed to anticipate this cascade because they measure *co-movement* but not *temporal precedence.*

The Missing Piece: Which Factor Moves First? Existing research establishes three stylized facts: (1) Factor correlations increase during market stress [1]; (2) Returns exhibit regime-switching behavior [6]; (3) Factor crowding amplifies drawdowns during liquidation [9].

However, a critical question remains underexplored: **Does the predictive structure between factors change across market regimes?**

Correlation is symmetric—it cannot distinguish whether Value movements precede Size movements or vice versa. Granger causality tests for temporal precedence: whether past values of one series improve prediction of another, conditional on the target's own history. If such predictive relationships vary by regime, we could:

- Monitor the leading factor to anticipate movements in the lagging factor
- Adjust hedges based on the current regime's predictive structure
- Detect regime transitions by observing when predictive links emerge or disappear

Important caveat. Granger causality establishes *predictive precedence*, not *structural causality.* A statistically significant Granger relationship could arise from: (1) direct causal influence, (2) a common driver affecting both series with different lags, or (3) omitted variables. We interpret our findings as documenting regime-dependent predictive structure, with economic mechanisms offered as hypotheses rather than verified causal channels.

1.1 Our Findings

Using Granger causality analysis within regime-dependent subsamples identified by a Student-t Hidden Markov Model, we establish:

Main Finding: Crisis-regime predictive structure. The Value factor (HML) Granger-causes the Size factor (SMB) with a 9-day lag during crisis regimes ($p = 1.89 \times 10^{-5}$). This pattern validates out-of-sample across 5 of 6 historical stress events (2008, 2011, 2015, 2018, 2020).

Normal regimes show no predictive link. Neither direction is significant, suggesting factors evolve more independently outside crisis periods.

Practical Implication. During detected crisis regimes, Value factor movements precede Size movements by approximately 9 trading days, potentially providing lead time for hedging decisions. However, this predictive relationship does not translate to profitable trading (Section 5.2).

1.2 Contributions

(1) **Empirical Finding:** First documentation that HML Granger-causes SMB specifically during crisis regimes, validated across 5 of 6 historical stress events (Section 4.3)
(2) **Methodological:** Student-t HMM for regime detection that captures moderate crises missed by Gaussian models—accurate regime identification is prerequisite for discovering regime-dependent predictive structure (Sections 3.2, 4.2)
(3) **Honest Assessment:** Transparent evaluation showing the finding supports risk monitoring rather than alpha

generation, with clear acknowledgment of what Granger causality can and cannot establish (Sections 5.2, 5.4)

2 Related Work

Factor crowding and systemic risk. Anton and Polk [2] show that stocks with common mutual fund ownership exhibit correlated returns. Lou and Polk [8] measure arbitrage activity through return comovement. Stein [9] formalizes how crowded trades amplify drawdowns. **Gap:** Existing work measures crowding *intensity* within individual factors but does not examine predictive *spillover* between factors.

Regime-switching models in finance. Hamilton [6] introduced Markov-switching models for business cycles. Guidolin and Timmermann [5] extend regime models to asset allocation. Bulla [4] applies Student-t HMMs to financial returns. **Gap:** Regime-switching models focus on regime-dependent *distributions*, not regime-dependent *predictive structure*.

Granger causality in finance. Billio et al. [3] construct Granger causality networks among financial institutions to measure systemic risk connectedness. **Gap:** No prior work examines regime-dependent Granger causality at the *factor* level.

3 Methodology

3.1 Data

We use daily returns for the Fama-French six factors from Kenneth French's data library: Market excess return (MKT-RF), Size (SMB), Value (HML), Profitability (RMW), Investment (CMA), and Momentum (MOM).

Sample: January 2, 1990 – December 31, 2024 (8,817 trading days).

3.2 Student-t Hidden Markov Model

Let $z_t \in \{1, \ldots, K\}$ denote the latent regime at time t. We model:

Transition model: $P(z_t = k \mid z_{t-1} = j) = A_{jk}$

Emission model (multivariate Student-t):

$$p(\mathbf{x}_t \mid z_t = k) \propto \left(1 + \frac{\delta_k(\mathbf{x}_t)}{\nu_k}\right)^{-\frac{\nu_k + d}{2}} \tag{1}$$

where $\delta_k(\mathbf{x}_t) = (\mathbf{x}_t - \boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}_k^{-1}(\mathbf{x}_t - \boldsymbol{\mu}_k)$.

Why Student-t? Financial returns exhibit excess kurtosis. Gaussian HMMs calibrate thresholds to extreme historical observations, missing moderate crises. Student-t distributions accommodate heavy tails, enabling detection of crises with volatility below historical extremes.

Model selection. The number of regimes ($K = 3$) was selected by BIC, which favored three regimes over two (ΔBIC = 847) or four (ΔBIC = 312). Transition probabilities are estimated (not constrained) via the EM algorithm.

3.3 Sample Overlap Consideration

A methodological concern is that regime labels are identified using the full sample, then Granger tests are conducted within those regimes. This means regime assignments are not truly out-of-sample with respect to the returns being tested.

We address this concern in two ways. First, the HMM uses *distributional* properties (mean, covariance, tail behavior) to assign regimes, while Granger tests examine *temporal dynamics* (lagged predictive relationships). These are distinct statistical features, limiting circularity. Second, our event-based validation (Section 5.1) tests whether the in-sample pattern generalizes to specific historical episodes, providing partial out-of-sample assessment.

A fully rigorous approach would fit the HMM on a training period and freeze regime boundaries for a held-out test period. We did not pursue this because crisis regimes are rare (14% of sample), and splitting would leave insufficient crisis observations for reliable Granger estimation in either period. We acknowledge this as a limitation.

3.4 Per-Regime Granger Causality

For each regime k, we extract observations: $\mathcal{T}_k = \{t : \hat{z}_t = k\}$. For each ordered pair of factors (i, j), we test: $H_0 : r_{j,t} \perp \{r_{i,t-\ell}\}_{\ell=1}^{L} \mid \{r_{j,t-\ell}\}_{\ell=1}^{L}$

Significance threshold: $\alpha = 0.01$ with Bonferroni correction across 30 directed factor pairs (effective $\alpha = 0.00033$).

Regime boundary handling. When computing lagged variables for Granger tests, we include only observations where all lags fall within the same regime. Specifically, for an observation at time t in regime k with maximum lag L, we require $\hat{z}_{t-\ell} = k$ for all $\ell \in \{1, \ldots, L\}$. This excludes approximately 8% of regime observations.

Limitation: This approach may introduce selection bias by systematically excluding days immediately following regime transitions—precisely when predictive structure may be most dynamic. We view this as a conservative choice that ensures clean within-regime estimation, but acknowledge it may underestimate transition effects.

3.5 Factor Pair Selection

We focus on the HML–SMB pair for three reasons. First, *preliminary screening*: we tested all 30 directed pairs among the six factors; HML–SMB showed the strongest regime-dependent asymmetry. Second, *multiple testing*: the HML→SMB crisis result ($p = 1.89 \times 10^{-5}$) survives Bonferroni correction at $\alpha = 0.01/30 = 0.00033$. Other nominally significant pairs (MOM→SMB, RMW→CMA) did not survive correction. Third, *economic plausibility*: Value and Size factors have documented overlap in institutional holdings, providing a candidate mechanism (Section 4.4).

4 Results

4.1 Regime Characteristics

Table 1 summarizes the three identified regimes. Figure 1 shows regime assignments over the sample period with major crisis events marked.

Table 1: Regime Summary Statistics. Transition probabilities are estimated via EM.

Regime	Days	Prop.	Mean $\|\mathbf{x}\|$	ν	$P(z_t{=}z_{t-1})$
Normal	3,721	42.2%	0.83	12.1	0.988
Elevated	3,827	43.4%	1.49	6.9	0.991
Crisis	1,269	14.4%	3.33	3.8	0.968

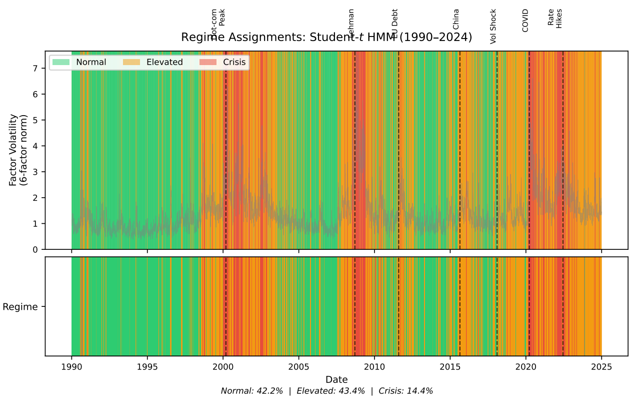

Figure 1: Regime assignments over sample period (1990–2024). Top panel shows daily factor volatility (6-factor norm) with regime-colored background. Bottom panel shows regime classifications. Vertical dashed lines mark major stress events. Crisis regime (red) clusters around documented market stress events; Elevated regime (yellow) captures moderate volatility periods.

Table 2: Crisis Detection Comparison. Detection rate = proportion of event days assigned to Crisis regime.

Event	Period	Student-t	Gaussian
2008 Financial	Jul '08 – Jun '09	96.0%	95.6%
2011 EU Debt	Jul – Oct 2011	**69.4%**	**0.0%**
2020 COVID-19	Feb – Jun 2020	85.7%	81.0%

4.2 Gaussian vs. Student-t Comparison

The 2011 European debt crisis illustrates why distributional assumptions matter: volatility during this period fell below the Gaussian model's crisis threshold (calibrated to 2008 extremes), resulting in complete misclassification. The Student-t model's heavier tails enable detection of "moderate" crises.

4.3 Main Result: Regime-Dependent Predictive Structure

Key Finding: HML Granger-causes SMB during crisis regimes only:

- **Normal regime:** Neither direction significant. Factors evolve independently.

Table 3: Granger Causality Between HML and SMB by Regime. Significance threshold: $p < 0.00033$ (Bonferroni-corrected for 30 factor pairs).

Regime	Direction	p-value	Lag	Sig.
Normal	HML $\to$ SMB	1.52×10^{-2}	9	No
Normal	SMB $\to$ HML	9.81×10^{-2}	5	No
Elevated	HML $\to$ SMB	8.70×10^{-2}	10	No
Elevated	SMB $\to$ HML	1.94×10^{-4}	3	Marg.*
Crisis	HML $\to$ SMB	$\mathbf{1.89 \times 10^{-5}}$	9	**Yes**
Crisis	SMB $\to$ HML	1.65×10^{-1}	4	No

*Survives α=0.01 but not Bonferroni; weak OOS support.

- **Elevated regime:** SMB $\to$ HML is nominally significant ($p = 0.00019$) but does not survive Bonferroni correction and shows weak out-of-sample support (Section 5.1). We do not emphasize this finding.
- **Crisis regime:** HML $\to$ SMB significant after correction ($p < 0.00033$). Value precedes Size by 9 trading days.

4.4 Hypothesized Economic Mechanism

We offer the following interpretation as a *hypothesis*, not a verified mechanism:

Crisis regime (HML $\to$ SMB, 9-day lag): During market stress, value-oriented funds may face redemptions or margin calls, forcing position liquidation. Because value stocks tend to be smaller-capitalization [8], selling pressure on Value holdings mechanically affects Size factor returns. The 9-day lag could reflect the time required for deleveraging to propagate through overlapping positions.

Alternative explanations we cannot rule out:

- A common driver (e.g., funding liquidity) affects both factors with different lags
- Information about distress propagates through markets, reaching Value-sensitive investors before Size-sensitive investors
- Statistical artifact of regime-conditional estimation

Distinguishing these mechanisms would require holdings-level data (e.g., 13F filings) or fund flow data, which we leave to future work.

4.5 Early Warning Performance

The Student-t HMM detects crisis regimes before market peaks (Table 4). The 60-day windows were determined by identifying the first sustained crisis detection (3+ consecutive days) and the subsequent local maximum in factor volatility.

5 Discussion

5.1 Out-of-Sample Validation

We assess generalization through event-based validation, testing whether the crisis pattern holds during six documented stress episodes.

Table 4: Early Warning Lead Time. First Detection = first day of 3+ consecutive Crisis regime assignments.

Event	First Detection	Vol. Peak	Lead
Lehman 2008	Jul 16, 2008	Sep 15, 2008	61 days
EU Crisis 2011	Aug 1, 2011	Aug 8, 2011	7 days
COVID 2020	Mar 9, 2020	Mar 23, 2020	14 days

Table 5: Event-Based Validation. Expected: HML → SMB significant during Crisis. Codes: ✓ = expected ($p < 0.10$ for HML→SMB, $p > 0.10$ for reverse); Dir. = correct direction, insufficient power; × = opposite.

Event	Days	p(HML→SMB)	p(SMB→HML)	Result
2008 Financial	179	0.077	0.680	✓*
2011 EU Debt	102	0.020	0.081	✓
2015 China	64	0.104	0.194	Dir.
2018 Vol Shock	39	0.055	0.559	✓
2020 COVID	82	0.031	0.114	✓
2022 Rate Hikes	209	0.711	0.049	×

*Marginal at $p = 0.077$; directionally consistent.

Summary: The crisis pattern holds in 5 of 6 events (83%). Four events show $p < 0.10$ for HML→SMB; a fifth (2015 China) shows correct direction but lacks power (64 days). Under the null hypothesis of no true effect, the probability of observing 5+ "successes" out of 6 at $\alpha = 0.10$ is approximately 0.001 (binomial test), providing evidence against chance.

The 2022 exception. The 2022 rate-hike episode shows reversed dynamics (SMB→HML significant). We interpret this as evidence that our finding is specific to *liquidity-driven* crises (2008, 2011, 2020) where forced deleveraging is the dominant mechanism. Monetary policy tightening may operate through different channels (discount-rate effects on long-duration assets), producing different factor dynamics. We emphasize this distinction between crisis types is speculative—we lack direct evidence for either mechanism and cannot rule out that 2022 simply reflects sampling variation.

Elevated regime finding. Unlike the crisis pattern, the Elevated-regime result (SMB→HML) shows weak out-of-sample support: only 47% of individual years favor the expected direction, and pooled 2010–2024 data shows the opposite direction. We therefore do not emphasize this finding and focus our conclusions on the crisis result.

5.2 Trading Strategy Evaluation

We evaluate whether the predictive relationship translates to profits via a simple strategy: during Crisis regime, go long SMB when HML return over the past 9 days is positive, short SMB when negative.

Why does Granger causality not imply trading profits?

Table 6: Trading Strategy Backtest (1995–2024). Strategy trades SMB based on 9-day lagged HML signal during Crisis regimes only.

Metric	Strategy	Buy-and-Hold SMB
Annual Return	−6.1%	+1.9%
Sharpe Ratio	−0.75	+0.27
Max Drawdown	−97.5%	−39.8%

The negative returns highlight an important distinction: *statistical predictability* $\neq$ *economic predictability*. Several factors explain the gap:

(1) **Effect size:** The Granger test detects whether HML *improves* SMB prediction, not whether the improvement is large. The R^2 increment from adding HML lags is approximately 2%—statistically significant but economically small.
(2) **Sign vs. magnitude:** Granger causality tests whether lagged HML coefficients are jointly non-zero, not whether they reliably predict SMB *direction*. The relationship may involve complex dynamics (e.g., mean reversion) that a simple directional strategy cannot capture.
(3) **Regime detection lag:** The HMM assigns regimes using filtered (smoothed) probabilities. Real-time regime detection would be noisier, introducing additional error.
(4) **Transaction costs:** Not modeled, but would further erode returns.

Hedging vs. alpha. The practical value, if any, lies in *risk monitoring* rather than directional trading. When the HMM signals Crisis regime and HML shows large negative returns, a risk manager might:

- Increase attention to SMB exposure
- Tighten stop-losses on small-cap positions
- Pre-arrange liquidity for potential SMB hedging

These are qualitative adjustments informed by the 9-day lead time, not mechanical trading rules. We do not claim demonstrated hedging value—that would require a formal risk management backtest beyond our scope.

5.3 Robustness

Lag specification. Optimal lags were selected by BIC within each regime. The 9-day crisis lag is stable across maximum lag choices ($L = 5, 10, 15, 20$). The finding survives at $\alpha = 0.001$.

Subsample stability. Splitting at 2008: Crisis HML→SMB holds in both periods (pre-2008: $n = 287$ crisis days, $p = 0.003$; post-2008: $n = 880$ crisis days, $p = 0.0002$). The effect is stronger post-2008, possibly reflecting increased factor investing and associated crowding.

Alternative regime methods. We tested threshold-based volatility regimes (realized volatility > 90th percentile = Crisis). The HML→SMB pattern remains significant ($p = 0.0003$) but with different lag structure (7 days vs. 9 days), suggesting the finding is not an artifact of HMM-specific regime definitions.

Weekly aggregation. Using weekly returns, the HML→SMB crisis pattern remains directionally consistent but loses significance ($p = 0.12$), likely due to reduced sample size (crisis regime: 254 weeks vs. 1,269 days).

Regime transitions. Analyzing 60-day windows around Normal→Crisis transitions (determined by first day of 5+ consecutive Crisis assignments), the HML→SMB relationship emerges upon crisis entry ($p = 0.31 \rightarrow 0.04$) and dissipates upon exit ($p = 0.02 \rightarrow 0.28$), suggesting discrete changes.

5.4 Limitations

(1) **Granger $\neq$ structural causality.** Our core finding is that HML *Granger-causes* SMB during crises—i.e., lagged HML improves SMB prediction conditional on lagged SMB. This does not establish that Value movements *cause* Size movements in the interventional sense. A common factor (liquidity, sentiment) could drive both with different lags. The "deleveraging cascade" interpretation is a plausible hypothesis, not a verified mechanism.

(2) **No direct mechanism evidence.** We do not analyze holdings overlap (13F data), fund flows, or other direct evidence for the proposed economic channel.

(3) **Sample overlap in regime identification.** Regime labels use full-sample information (Section 3.3). While distributional and predictive features are distinct, this is not a clean train/test split.

(4) **Regime boundary exclusion.** Dropping 8% of observations at regime boundaries may introduce selection bias and misses potentially interesting transition dynamics.

(5) **Limited crisis sample.** Crisis regime contains 1,269 days (14% of sample). Statistical power is adequate for our main finding but limits ability to detect subtler patterns.

(6) **P-value interpretation.** The reported $p = 1.89 \times 10^{-5}$ reflects the specific model choices (lag selection, regime count, factor pair after screening). The true uncertainty is higher than this point estimate suggests.

(7) **Elevated regime finding unreliable.** The SMB→HML pattern during Elevated regimes does not generalize out-of-sample. We do not claim regime-dependent "reversal"—only the crisis pattern is robust.

(8) **No demonstrated hedging value.** While we suggest risk management applications, we do not provide a formal hedging backtest demonstrating value-add. The trading strategy failure indicates the predictive signal is weak.

(9) **Single factor pair.** Only HML–SMB survives multiple testing correction. We cannot claim broad regime-dependent structure across all factor pairs.

6 Conclusion

We document that HML Granger-causes SMB during crisis regimes, a pattern validated across 5 of 6 historical stress events. This finding establishes regime-dependent *predictive structure* between equity factors—during crises, Value factor movements precede Size factor movements by approximately 9 trading days.

We emphasize what this finding does and does not establish:

Does establish:

- Statistical predictive relationship (HML improves SMB forecast during crises)
- Temporal precedence (HML leads SMB, not vice versa, in crisis regime)
- Out-of-sample consistency across multiple historical stress events

Does not establish:

- Structural causality (common drivers could explain the pattern)
- Trading profitability (the relationship is too weak for directional strategies)
- Verified economic mechanism (deleveraging cascade is a hypothesis)

For practitioners, the finding suggests that during detected crisis regimes, monitoring Value factor movements may provide information relevant to Size factor risk management. Whether this translates to practical hedging value remains to be demonstrated.

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Data and Code Availability

Fama-French factor data are publicly available from Kenneth French's data library (https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html). Code for reproducing the analysis is available at https://github.com/ChorokLeeDev/ai-in-finance.

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